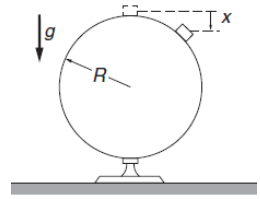
PHY101: Introduction to Physics-1

Monsoon, 2017

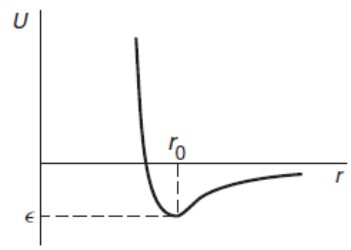
Tutorial-11

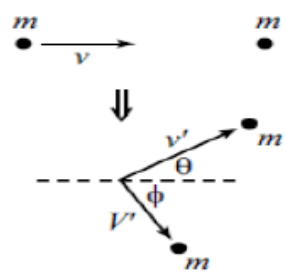
1. A small block slides from rest from the top of a frictionless sphere of radius R, as shown in the figure below. How far below the top x does it lose contact with the sphere? Consider that the sphere does not move.



2. A commonly used potential energy function, *U(r)* to describe the interaction between two atoms is the Lennard-Jones potential given by the equation below and shown in the figure.

$$U(r) = \varepsilon[(\frac{r_0}{r})^{12} - 2(\frac{r_0}{r})^6]$$



(*a*) Find the position of the potential minimum and its value.
(*b*) Near the minimum the atoms execute simple harmonic motion. Find the frequency of oscillation.

3. A billiard ball with speed *v* approaches an identical stationary one. The balls bounce off each other elastically, in such a way that the incoming one gets deflected by an angle θ. What are the final speeds of the balls? What is the angle, φ, at which the stationary ball is ejected?